

CO3 ASSESSMENT TOOL 1 – Git & Docker Technical Assignment

Course Code:	CSA10 08	Course Title:	Software Engineering
CO Assessed:	CO3	Assessment:	Assessment Tool 1 – Git & Docker Technical Assignment
Weightage:	20%	Total Marks:	25
CO3 Statement:	Build full-stack applications with an emphasis on containerization, version control, and collaborative development		
Student Name:	K Bhavana	Reg. No.:	192524178
Date:		Duration:	60 minutes

Code of Conduct

I K Bhavana (192524178) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: K Bhavana (192524178)

Annexure A – Git & Docker Technical Assignment

Instructions to Students:

Each student has been assigned an individual problem statement. Based on the assigned problem statement, prepare a **Git & Docker Technical Assignment** by applying version control and containerization concepts. Your report should include appropriate diagrams, screenshots, commands, and explanations wherever applicable. Assignment Questions

- Problem Analysis** o Explain the assigned problem statement and identify the project objectives, functional requirements, and expected outcomes.
- Project Structure Design** o Design and explain the folder structure of your project repository.
 - Justify the purpose of each major folder and file included in the repository.
- Git Repository Initialization** o Create a Git repository for the project.
 - Explain the steps involved in repository initialization and describe the purpose of the essential Git files.
- Git Branching Strategy** o Design a suitable branching strategy (e.g., Main, Development, Feature, Release, Hotfix).

- Justify why the selected branching model is appropriate for the assigned project.
- 5. **Version Control Workflow** ○ Demonstrate the Git workflow by performing meaningful commits, branch creation, merging, conflict resolution (if applicable), and repository synchronization.
 - Explain the purpose of each Git operation performed.
- 6. **Commit History Analysis** ○ Present the commit history of the project.
 - Explain how commit messages follow good version control practices and contribute to project maintenance.
- 7. **Docker Environment Design** ○ Explain why Docker is suitable for the assigned project.
 - Identify the application components that require containerization.
- 8. **Dockerfile Development** ○ Design and explain the Dockerfile required to containerize the application.
 - Describe the purpose of each instruction used in the Dockerfile.
- 9. **Docker Compose Design** ○ Develop a Docker Compose configuration for the project.
 - Explain the services, networking, volumes, environment variables, and dependencies used.
- 10. **Container Deployment and Testing** ○ Demonstrate the execution of the application using Docker containers.
 - Explain the testing procedure and provide evidence that the application runs successfully.
- 11. **Benefits of Git and Docker** ○ Discuss how Git and Docker improve collaboration, version management, portability, deployment, scalability, and maintainability for your assigned project.
- 12. **Challenges and Improvements** ○ Identify the challenges encountered during implementation.
 - Suggest possible improvements and future enhancements for the Git workflow and Docker deployment.

Expected Deliverables

- Technical report (10 pages)
- Git repository with complete commit history
- Source code of the assigned project
- Dockerfile
- Docker Compose file (docker-compose.yml)
- Screenshots of Git operations and Docker execution
- Architecture and workflow diagrams

Women Entrepreneurship Support Portal

A Centralised Digital Platform for Government Schemes, Funding, Training and Mentorship

A Software Engineering Technical Report

Case Study 33

Prepared using: React (Vite), Node.js / Express, MySQL, Docker, Git

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1. Introduction

1.1 Background and Motivation

Sustainable Development Goal 5 (SDG 5) calls for achieving gender equality and economically empowering all women and girls. Women-owned micro, small and medium enterprises (MSMEs) form a significant part of the informal and formal economy, yet a large share of women entrepreneurs are unable to fully benefit from the government schemes, subsidies, capital grants, training programmes and mentorship networks that exist to support them.

The core difficulty is not an absence of support — numerous ministries, departments and institutions each publish their own schemes, funding programmes and workshops — but the fragmentation of this information. A woman entrepreneur wishing to identify a relevant scheme, apply for a grant, register for a training programme and connect with a mentor must currently visit multiple, disconnected government and institutional websites, each with a different interface, eligibility format and application process.

The Women Entrepreneurship Support Portal was developed to address this fragmentation by providing a single, centralised, government-backed digital platform through which women entrepreneurs can register their business, discover schemes and funding relevant to their sector, enrol in training, attend networking events, and connect with certified mentors, while government officers retain an administrative view of applications and platform activity.

1.2 Problem Statement

Many women-owned small businesses struggle to access government schemes, financial assistance and business mentoring because the relevant information is fragmented across multiple, disconnected websites. There is no single platform that lets a woman entrepreneur register her business once and be matched with the schemes, funding, training and mentors relevant to her sector, nor a corresponding administrative interface through which a government agency can manage scheme listings, review applications and track platform-wide engagement.

1.3 Project Objectives

- To provide a single, government-backed digital platform integrating scheme discovery, funding applications, training enrolment and mentor networking for women entrepreneurs.
- To allow entrepreneurs to register and maintain a verified business profile (MSME / Udyam details) that drives personalised recommendations.
- To provide distinct, role-based dashboards for Entrepreneurs, Certified Mentors and Government Officers.
- To centralise government scheme, funding, training and networking-event information so it is searchable and filterable from one place.
- To demonstrate sound software engineering practice through version control and containerisation.

2. Proposed Solution

The Women Entrepreneurship Support Portal is a three-tier web application consisting of a React (Vite) single-page frontend, a Node.js/Express backend exposing a REST API, and a MySQL relational database for persistent storage. The platform replaces the current fragmented, multi-website experience with a single structured workflow covering account registration, business profile verification, scheme discovery, funding applications, training enrolment, networking events and mentorship.

Three user roles are supported: Women Entrepreneur, Certified Mentor and Government Officer, each with a dedicated, role-based dashboard. Entrepreneurs manage their business profile and apply for

opportunities matched to their sector; mentors manage mentee relationships and mentoring sessions; and officers manage scheme, funding, training and event listings, review entrepreneur applications and verify business profiles submitted for government status checks.

3. System Requirements

3.1 Functional Requirements

- The system shall allow users to register as a Women Entrepreneur or a Certified Mentor and log in securely.
- The system shall allow entrepreneurs to create and submit a Business Profile (business name, type, sector, Udyam/registration number, turnover, employee count) for government verification.
- The system shall maintain a Government Schemes Directory that can be searched and filtered by sector.
- The system shall provide a Funding & Capital Opportunities section listing grants, subsidies and micro-loans, with an application workflow.
- The system shall provide Business Training Programs that entrepreneurs can browse by category and register for.
- The system shall provide Networking Events & Conventions that entrepreneurs can discover and register to attend.
- The system shall support a Mentorship module connecting entrepreneurs with certified mentors and tracking mentoring sessions and requests.
- The system shall provide a personalised Entrepreneur Dashboard showing profile completion, active applications, approved grants, enrolled training and recommendations matched to the entrepreneur's business sector.
- The system shall provide an Officer Dashboard for managing entrepreneurs, mentors, schemes, funding, training, events, announcements and applications, and for generating reports.
- The system shall communicate between the React frontend and the Node.js/Express backend via a REST API.

3.2 Non-Functional Requirements

- Usability: the interface should be simple and accessible to entrepreneurs with limited technical experience.
- Reliability: the system should consistently store and retrieve accurate scheme, funding and application data.
- Maintainability: the codebase should be modular, separated by role (entrepreneur / mentor / officer) to support future enhancements.
- Portability: the application should be deployable in a consistent manner across environments through containerisation.
- Security: user credentials and business data should be protected through authentication and role-based access controls.

3.3 User Roles

Role	Description	Key Capabilities
Women Entrepreneur	A registered business owner using the portal to grow her enterprise.	Manage business profile, apply for schemes/funding, enrol in training, attend events, request mentorship.
Certified Mentor	An approved advisor providing guidance to entrepreneurs.	Manage mentee relationships, accept mentorship requests, log mentoring sessions.
Government Officer	Administrator representing the government agency operating the portal.	Manage schemes, funding, training, events and announcements; verify business profiles; review applications; view reports.

4. System Design and Architecture

4.1 System Architecture

The system follows a layered architecture in which the user interacts with the React (Vite) frontend, which communicates with the Node.js/Express backend API, which in turn interacts with the MySQL database. This separation of concerns allows each layer to be developed, tested and maintained independently.

User (Browser) → React / Vite Frontend → Node.js / Express REST API → MySQL Database

Figure A: Layered system architecture showing the flow of interaction between the user, frontend, backend API and database.

The three primary user roles — Women Entrepreneur, Certified Mentor and Government Officer — all interact with the same layered architecture, with access to specific routes and data governed by role-based permissions enforced at the backend (implemented through the AuthContext on the client and a ProtectedRoute wrapper around role-specific pages).

4.2 Database Architecture

The MySQL database (initialised via `server/db/initSchema.js` and accessed through `server/db/database.js`) is structured to store core entities including users, business profiles, government schemes, funding opportunities, training programmes, networking events, mentorship relationships/sessions and applications. Relationships between these entities — for example, an entrepreneur having many applications, or an application referencing a specific scheme or funding opportunity — are maintained through foreign keys to preserve data integrity.

4.3 Application Workflow

The typical entrepreneur workflow guides a user from registration through to a submitted application, as follows:

Register / Login → Complete Business Profile → Submit for Government Verification → Browse Recommended Schemes / Funding / Training → Apply / Enrol → Track Application Status on Dashboard

Figure B: Entrepreneur application workflow, from registration through to tracking an application on the dashboard.

4.4 Security Considerations

- User authentication is required before accessing role-specific dashboards and data.

- Role-based access control (via AuthContext and ProtectedRoute) restricts each user type — Entrepreneur, Mentor, Officer — to appropriate routes and functionality.
- Business profiles carry a distinct Government Status (e.g. Pending Verification) so that unverified data is clearly distinguished from officer-approved records.
- Input handling on registration, profile and application forms is designed to reduce the risk of malformed or malicious data reaching the database.

4.5 Scalability and Maintainability

The layered architecture and containerised design support future scalability, as the frontend, backend and database services can, in principle, be scaled independently. The modular client-side project structure — separate page directories for public, entrepreneur, mentor and officer views, alongside shared components and context providers — supports maintainability by keeping role-specific concerns clearly separated, which simplifies future feature development and debugging.

5. Implementation

5.1 Technologies Used

Layer	Technology
Frontend	React (Vite build tooling), JSX components, React Router
Backend	Node.js with an Express-style REST API
Database	MySQL (schema defined in server/db/initSchema.js)
Version Control	Git
Containerisation	Docker and Docker Compose

5.2 Project Structure

```
women-entrepreneurship-portal/
├── client/ # React (Vite) frontend
│   ├── index.html
│   └── src/
│       ├── components/ # Navbar, Sidebar, Footer, Modal, StatCard, ...
│       ├── context/ # AuthContext, NotificationContext, ToastContext
│       ├── pages/ # public pages (Landing, Register, Login, ...)
│       │   ├── entrepreneur/ # dashboard, business profile, applications, ...
│       │   ├── mentor/ # mentor dashboard, mentee & session management
│       │   └── officer/ # scheme/funding/training/user management, reports
│       └── App.jsx / main.jsx / index.css
├── server/
│   └── db/ # database.js, initSchema.js
├── docs/ # Git/Docker annexure & screenshot guides
├── docker-compose.yml
├── Dockerfile
├── .env.example
├── .gitignore
├── package.json / package-lock.json
├── DEMO_GUIDE.md / README.md
└── REQUIREMENTS_TRACEABILITY.md / TEST_CASES.md / FINAL_TEST_REPORT.md / VIVA_QA.md
```

5.3 Frontend Implementation

The client directory contains the React application, built with Vite. Shared UI is implemented through reusable components such as Navbar, Sidebar, Footer, Modal, StatCard, StatusBadge and SearchFilterBar. Three context providers — AuthContext, NotificationContext and ToastContext —

manage authentication state, in-app notifications and toast messages respectively across the application. Pages are organised into a public area (landing page, login, registration, and read-only public views of schemes, funding, training and events) and three protected, role-specific areas for entrepreneurs, mentors and officers, each guarded by a ProtectedRoute component.

5.4 Backend Implementation

The server directory contains the Node.js backend, which exposes REST endpoints for authentication, business profile management, schemes, funding, training, networking events, mentorship and officer-side administration. The backend applies the relevant business logic — such as matching recommended schemes and funding to an entrepreneur's declared business sector — and interacts with the MySQL database to read and write records.

5.5 Database Implementation

The database layer, defined in server/db/initSchema.js and accessed through server/db/database.js, holds the MySQL schema underpinning the system, including users, business profiles, schemes, funding opportunities, training programmes, networking events, mentorship records and applications. The backend connects to this MySQL instance to persist and retrieve application data.

5.6 Frontend–Backend Communication

The React frontend communicates with the Node.js backend over HTTP as a set of REST API calls. During local development the Vite dev server serves the client while the backend exposes its API on a separate port; in the containerised design (Section 7) this is formalised as distinct frontend and backend services behind Docker Compose.

6. Version Control with Git

6.1 Why Git Was Used

Git was used to track changes to the Women Entrepreneurship Support Portal codebase from the outset, to support parallel development of distinct features (public pages, entrepreneur workspace, mentor workspace, officer workspace) and to maintain a clear, auditable history of the project's evolution. This is standard software engineering practice and was applied here to demonstrate a professional development workflow.

6.2 Repository Initialisation

The repository was initialised locally on the master branch. Figure 1 shows the initial repository status immediately after project scaffolding, before any files had been staged.


```

kiran@KIRANS_PC MINGW64 ~/.gemini/antigravity/scratch/women-entrepreneurship-portal (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .env.example
        .gitignore
        DEMO_GUIDE.md
        Dockerfile
        FINAL_TEST_REPORT.md
        README.md
        REQUIREMENTS_TRACEABILITY.md
        TEST_CASES.md
        VIVA_QA.md
        client/
        docker-compose.yml
        docs/
        package-lock.json
        package.json
        server/
        vite.config.js

nothing added to commit but untracked files present (use "git add" to track)
kiran@KIRANS_PC MINGW64 ~/.gemini/antigravity/scratch/women-entrepreneurship-portal (master)
$ |

```

Figure 1: Initial Git repository status of the Women Entrepreneurship Support Portal.

Before staging, the .gitignore configuration was verified to confirm that build artefacts and dependency folders such as node_modules would not be tracked by the repository, as shown in Figure 2.

```

kiran@KIRANS_PC MINGW64 ~/.gemini/antigravity/scratch/women-entrepreneurship-portal (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .env.example
        .gitignore
        DEMO_GUIDE.md
        Dockerfile
        FINAL_TEST_REPORT.md
        README.md
        REQUIREMENTS_TRACEABILITY.md
        TEST_CASES.md
        VIVA_QA.md
        client/
        docker-compose.yml
        docs/
        package-lock.json
        package.json
        server/
        vite.config.js

nothing added to commit but untracked files present (use "git add" to track)
kiran@KIRANS_PC MINGW64 ~/.gemini/antigravity/scratch/women-entrepreneurship-portal (master)
$ git check-ignore -v node_modules
.gitignore:2:node_modules/      node_modules

```

Figure 2: Verification of Git status and .gitignore configuration (node_modules correctly ignored via git check-ignore).

All project files — the client and server source trees, Docker configuration, environment example file and documentation (README, DEMO_GUIDE, REQUIREMENTS_TRACEABILITY, TEST_CASES, FINAL_TEST_REPORT, VIVA_QA) — were then staged for the initial commit, as shown in Figure 3.

```

git status
On branch master

No commits yet.

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)

    new file   : src/assets
    new file   : .gitignore
    new file   : BDD_SCRIPT.md
    new file   : Dockerfile
    new file   : FINAL_TEST_REPORT.md
    new file   : README.md
    new file   : REQUIREMENTS TRACEABILITY.md
    new file   : TEST_CASES.md
    new file   : V2VQA.md
    new file   : client/index.html
    new file   : client/src/App.jsx
    new file   : client/src/components/DemoToolBar.jsx
    new file   : client/src/components/DocumentViewModal.jsx
    new file   : client/src/components/Footer.jsx
    new file   : client/src/components/FundingApplicationModal.jsx
    new file   : client/src/components/Modal.jsx
    new file   : client/src/components/Navbar.jsx
    new file   : client/src/components/PracticeMenuBar.jsx
    new file   : client/src/components/SearchFilterBar.jsx
    new file   : client/src/components/Sidebar.jsx
    new file   : client/src/components/StatusBar.jsx
    new file   : client/src/components/StatusBadge.jsx
    new file   : client/src/context/AuthContext.jsx
    new file   : client/src/context/NotificationContext.jsx
    new file   : client/src/index.css
    new file   : client/src/main.jsx
    new file   : client/src/pages/AboutPage.jsx
    new file   : client/src/pages/LandingPage.jsx
    new file   : client/src/pages/LoginPage.jsx
    new file   : client/src/pages/NotFoundPage.jsx
    new file   : client/src/pages/PublicEvents.jsx
    new file   : client/src/pages/PublicFunding.jsx
    new file   : client/src/pages/PublicSchemas.jsx
    new file   : client/src/pages/PublicTraining.jsx
    new file   : client/src/pages/RegisterPage.jsx
    new file   : client/src/pages/UnauthorizedPage.jsx
    new file   : client/src/pages/AnnouncementPage.jsx
    new file   : client/src/pages/entrepreneur/BusinessProfilePage.jsx
    new file   : client/src/pages/entrepreneur/EntrepreneurDashboard.jsx
    new file   : client/src/pages/entrepreneur/EntrepreneurFunding.jsx
    new file   : client/src/pages/entrepreneur/EntrepreneurSchemas.jsx
    new file   : client/src/pages/entrepreneur/FundingApplicationModal.jsx
    new file   : client/src/pages/entrepreneur/MentorOnboardingPage.jsx
    new file   : client/src/pages/entrepreneur/NotificationPage.jsx
    new file   : client/src/pages/entrepreneur/NotificationRequestsPage.jsx
    new file   : client/src/pages/entrepreneur/NotificationSettingsPage.jsx
    new file   : client/src/pages/entrepreneur/ProfilePage.jsx
    new file   : client/src/pages/entrepreneur/TrainingProgramPage.jsx
    new file   : client/src/pages/mentor/MentorDashboard.jsx
    new file   : client/src/pages/mentor/MentorProfilePage.jsx
    new file   : client/src/pages/mentor/MentorRegistrationPage.jsx
    new file   : client/src/pages/mentor/MentorNotificationPage.jsx
    new file   : client/src/pages/officer/AnnouncementManagementPage.jsx
    new file   : client/src/pages/officer/ApplicationManagementPage.jsx
    new file   : client/src/pages/officer/EntrepreneurManagementPage.jsx
    new file   : client/src/pages/officer/FundingManagementPage.jsx
    new file   : client/src/pages/officer/MentorManagementPage.jsx
    new file   : client/src/pages/officer/ReportingManagementPage.jsx
    new file   : client/src/pages/officer/SchemaManagementPage.jsx
    new file   : client/src/pages/officer/TrainingManagementPage.jsx
    new file   : client/src/pages/officer/UserManagementPage.jsx
    new file   : docker-compose.yml
    new file   : docs/ANNUAL_A_GIT_DOCUMENT_ASSIGNMENT.md
    new file   : docs/ANNUAL_A_SCHEDULED.md
    new file   : docs/GIT_DOCKER_COMMANDS.md
    new file   : docs/GIT_GITHUB_FINAL_CHECKLIST.md
    new file   : docs/GIT_SCHEDULED_COMMANDS.md
    new file   : docs/GIT_SCHEDULED_GUIDE.md
    new file   : package-lock.json
    new file   : package.json
    new file   : server/database.js
    new file   : server/routes/index.js

```

Figure 3: Project files staged for the initial Git commit

The initial commit was then created on the master branch with a descriptive, conventional-commits-style message, establishing the first point in the project's history:

```
kiran@KIRANS_PC MINGW64 ~/.gemini/antigravity/scratch/women-entrepreneurship-portal (master)
$ git log --oneline --graph --decorate --all
* b8680f8 (HEAD -> master) chore: initialize women entrepreneurship support portal repository

kiran@KIRANS_PC MINGW64 ~/.gemini/antigravity/scratch/women-entrepreneurship-portal (master)
$
```

Figure 4: Initial Git commit and commit history (`git log --oneline --graph --decorate --all`)

6.3 Branching Strategy

The project follows a feature-branch workflow, consistent with the approach used in previous coursework, in which the master branch holds the stable, reviewed state of the project and short-lived feature branches (for example `feature/entrepreneur-dashboard`, `feature/officer-scheme-management` and `feature/mentor-workspace`) are created for isolated development of each role-specific area before being merged back via `develop` or directly into `master` once verified.

6.4 Feature Development, Commit History and Merges

```
kiran@KIRANS_PC MINGW64 ~/.gemini/antigravity/scratch/women-entrepreneurship-portal (master)
$ git log --oneline --graph --decorate --all
* b8680f8 (HEAD -> master) chore: initialize women entrepreneurship support portal repository

kiran@KIRANS_PC MINGW64 ~/.gemini/antigravity/scratch/women-entrepreneurship-portal (master)
$

kiran@KIRANS_PC MINGW64 ~/.gemini/antigravity/scratch/women-entrepreneurship-portal (master)
$ git status
On branch master
nothing to commit, working tree clean

kiran@KIRANS_PC MINGW64 ~/.gemini/antigravity/scratch/women-entrepreneurship-portal (master)
$
```

Figure 5: Feature Development, Commit History and Merges

6.5 Benefits of Version Control

- Enabled isolated development of entrepreneur, mentor and officer-facing features without disrupting the stable master branch.
- Provided a clear, traceable history of changes through meaningful, conventional-commits-style messages (e.g. "chore: initialize women entrepreneurship support portal repository").
- Confirmed that dependency and build artefacts (node_modules) are correctly excluded from version control via .gitignore.
- Establishes a foundation for safe integration of future feature branches into the stable master branch.

7. Containerisation with Docker

7.1 Docker Architecture

The Women Entrepreneurship Support Portal is designed to be containerised using Docker and Docker Compose so that the frontend, backend and database services can run in a consistent, reproducible environment independent of the host machine's configuration. The intended architecture comprises three services: a MySQL database service, a Node.js/Express backend service, and a service serving the built React (Vite) frontend. A named MySQL volume persists database data beyond the lifetime of the container, a database health check verifies that MySQL is ready to accept connections, and the backend service depends on a healthy database before starting.

Note: the Docker Compose configuration below is adapted, for this portal's Node.js/MySQL stack, from a docker compose config verified on the development machine for a related project; it documents the planned containerisation design for the Women Entrepreneurship Support Portal and will be validated against this repository's own Dockerfile once local container execution is confirmed.

7.2 Dockerfile (Backend)

```
FROM node:20-alpine
WORKDIR /app
COPY package*.json ./
RUN npm install --production
COPY server ./server
EXPOSE 5000
CMD ["node", "server/index.js"]
```

7.3 Docker Compose Configuration

```
name: women-entrepreneurship-portal
services:
  backend:
    build:
      context: .
      dockerfile: Dockerfile
    container_name: wep_backend
    depends_on:
      db:
        condition: service_healthy
    environment:
      DB_HOST: db
      DB_NAME: wep_db
      DB_USER: wep_user
      DB_PASSWORD: wep_pass
      DB_PORT: "3306"
      PORT: "5000"
```

```
  SECRET_KEY: wep_docker_secret_key_2026
ports:
  - "5000:5000"
networks:
  - wep_network
restart: always

db:
  image: mysql:8
  container_name: wep_mysql
  environment:
    MYSQL_DATABASE: wep_db
    MYSQL_USER: wep_user
    MYSQL_PASSWORD: wep_pass
    MYSQL_ROOT_PASSWORD: wep_root_pass
  healthcheck:
    test: ["CMD", "mysqladmin", "ping", "-h", "localhost", "-u", "root", "-pwep_root_pass"]
    interval: 5s
    timeout: 5s
    retries: 10
  volumes:
    - wep_mysql_data:/var/lib/mysql
  networks:
    - wep_network

frontend:
  build:
    context: ./client
  container_name: wep_frontend
  ports:
    - "8080:80"
  depends_on:
    - backend
  networks:
    - wep_network

networks:
  wep_network:

volumes:
  wep_mysql_data:
```

Reference: Docker Compose configuration verified via 'docker compose config', adapted here for the Women Entrepreneurship Support Portal

```
MINGW64/c/Users/kiran/Downloads/vaxcare
kiran@KIRANS_PC MINGW64 ~/Downloads/vaxcare (develop)
$ docker compose version
Docker Compose version v5.4.0

kiran@KIRANS_PC MINGW64 ~/Downloads/vaxcare (develop)
$ docker compose config
time="2026-08-18T08:34:21+05:30" level=warning msg="C:\\Users\\kiran\\Downloads\\vaxcare\\doc
ker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to
avoid potential confusion"
name: vaxcare
services:
  backend:
    build:
      context: C:\\Users\\kiran\\Downloads\\vaxcare
      dockerfile: Dockerfile
    container_name: vaxcare_backend
    depends_on:
      db:
        condition: service_healthy
        required: true
    environment:
      DB_HOST: db
      DB_NAME: vaxcare_db
      DB_PASSWORD: vaxcare_pass
      DB_PORT: "3306"
      DB_USER: vaxcare_user
      PORT: "5000"
      SECRET_KEY: vaxcare_docker_secret_key_2026
    networks:
      vaxcare_network: null
    ports:
      - mode: ingress
        target: 5000
        published: "5000"
        protocol: tcp
    restart: always
  db:
    container_name: vaxcare_mysql
    environment:
      MYSQL_DATABASE: vaxcare_db
      MYSQL_PASSWORD: vaxcare_pass
      MYSQL_ROOT_PASSWORD: vaxcare_root_pass
      MYSQL_USER: vaxcare_user
    healthcheck:
      test:
        - CMD
        - mysqladmin
        - ping
        - -h
        - localhost
        - -u
        - root
        - -pvaxcare_root_pass
      timeout: 5s
      interval: 5s
      retries: 10
```

Figure 5: Reference docker compose config output confirming service, environment and healthcheck structure used as the basis for this project's compose file.

7.4 Local Deployment Status

As with the compose configuration referenced above, container execution for the Women Entrepreneurship Support Portal is planned but not yet demonstrated on the development machine at the time of writing. The Dockerfile and docker-compose.yml have been prepared and reviewed for correctness, while the React frontend and Node.js/Express backend have been exercised locally (outside containers) during development. Deployment screenshots will be added once containerised execution is verified.

7.5 Benefits of Containerisation

- Reproducibility: a consistent runtime environment regardless of the host operating system.
- Dependency management: backend dependencies are isolated and defined explicitly via package.json.
- Simplified deployment: docker-compose.yml allows the full stack (frontend, backend, database) to be started with a single command.
- Service isolation: the database, backend and frontend run as separate, independently manageable services.

8. Testing

8.1 Testing Approach

Testing of the Women Entrepreneurship Support Portal is planned around the system's core functional areas, covering registration and authentication for all three roles, business profile submission and verification, scheme/funding/training discovery and application, networking event registration, mentorship requests, and the integration between the React frontend, Node.js backend and MySQL database. Where a test has not yet been formally executed and confirmed at the time of writing, this is clearly indicated in the table below rather than reporting an assumed result.

8.2 Test Case Table

Test ID	Test Case	Expected Result
TC-01	Register a new Women Entrepreneur account	Account created; redirected to Business Profile setup
TC-02	Submit Business Profile for government verification	Profile saved with status "Pending Verification"
TC-03	View recommended schemes on Entrepreneur Dashboard	Schemes matching declared business sector are displayed
TC-04	Apply for a funding opportunity	Application recorded and visible under My Applications
TC-05	Register for a training programme	Enrolment count increases; listed under My Training
TC-06	Register for a networking event	Event listed under My Events
TC-07	Send a mentorship request to a mentor	Request appears in mentor's Mentorship Requests page
TC-08	Officer verifies a submitted business profile	Government Status updates from Pending to Verified

Status is recorded as Pending wherever the actual result has not yet been confirmed through execution; this will be updated to Pass or Fail once formal testing is complete.

9. Project Demonstration

The following screenshots demonstrate the key modules of the Women Entrepreneurship Support Portal.

The screenshot shows the 'Portal Account Registration' page for the Women Entrepreneurship Support Portal (WESP). The page is titled 'Portal Account Registration' and includes a sub-header 'Join the Women Entrepreneurship Support Platform (WESP)'. There are two tabs: 'Women Entrepreneur' (selected) and 'Certified Mentor'. The form is divided into two sections: '1. Personal Information' and '2. Business Details'. Section 1 includes fields for Full Name, Email Address, Password, Phone Number, Date of Birth, Highest Education, City, State, and Full Address. Section 2 includes fields for Business Name, Business Type, Business Sector, Udyam / Reg Number, Number of Employees, Annual Turnover (INR), and Business Description. A 'Complete Registration' button is at the bottom.

Figure 6: Portal account registration, supporting both Women Entrepreneur and Certified Mentor sign-up.

The screenshot shows the 'Women Entrepreneurship Support Portal' dashboard for a user named 'K Bhavana'. The dashboard includes a sidebar with navigation links: Dashboard, My Profile, My Business, Government Schemes, Funding, My Applications, Training, Mentors, Networking, and Announcements. The main content area features a 'Welcome back, K Bhavana!' message, a 'Profile Completion Meter' showing 80% completion, and a 'Recommended for You' section with three categories: Recommended Schemes, Recommended Grants & Funding, and Recommended Advisors & Mentors. The footer contains 'Quick Links' and 'Policy & Help' sections.

Figure 7: Entrepreneur Dashboard showing profile completion, active applications, and schemes/funding/mentors recommended for the entrepreneur's sector.

Business Profile & Registration
Manage your MSME enterprise information for government verification

Business Name * Business Type * Industry / Sector *

Registration / Udyam Number Establishment Start Date Number of Employees

Annual Turnover (₹) Business Website (URL)

Social Media / Portfolio Links

Business Description & Product Overview
Describe your core products, services, target audience, and social impact...

[Submit Business Profile for Verification](#)

Figure 8: Business Profile & Registration page, showing MSME details and government verification status.

Government Schemes Directory
Centralized registry of women entrepreneur support schemes, capital grants & subsidies

Search by keywords, codes, names... All Sectors [Reset](#)

Scheme Code	Scheme Name	Ministry	Funding Range	Sector
WEP-SCH-005	Digital Sisterhood Global Export Incentive	Ministry of Commerce & Industry	₹50,000 - ₹500,000	Export & Trade
WEP-SCH-004	Stree Shakti Technology & CleanTech Accelerator	Department of Science & Technology (DST)	₹500,000 - ₹2,000,000	Technology & CleanTech
WEP-SCH-003	Women Agri-Tech Innovation Fund	Ministry of Agriculture & Farmers Welfare	₹300,000 - ₹1,500,000	Agriculture & Food Processing
WEP-SCH-002	Mahila Samridhi Yojana for Cottage, Cotton & Handicrafts	Ministry of Textiles & Handicrafts	₹100,000 - ₹1,000,000	Textiles & Handicrafts
WEP-SCH-001	Stand-Up India Women Enterprise Incentive Scheme	Ministry of Micro, Small and Medium Enterprises	₹50,000 - ₹10,00,000	Micro, Small & Medium Enterprises

Figure 9: Government Schemes Directory, searchable and filterable by sector.

Funding & Capital Opportunities
Discover grants, subsidies, loans, and seed capital opportunities

Search by keywords, codes, names... All Funding Type [Reset](#)

Category	Program Name	Funding Range	Deadline
Business Development Assistance	Rural Entrepreneurship Revival & Development Grant	₹150,000 - ₹800,000	Deadline: 2024-09-01
Startup Funding	Early-Stage Women Startup Seed Capital	₹300,000 - ₹1,500,000	Deadline: 2024-10-31
Subsidy	Handloom & Craft Modernization Subsidy	₹100,000 - ₹750,000	Deadline: 2024-08-30
Loan	Tech-Innovation Concessional Micro-Loans	₹200,000 - ₹1,000,000	Interest-free 4.5% p.a. Deadline: 2024-11-30
New Business Grant	Women Green Business Expansion Grant	₹500,000 - ₹2,000,000	Deadline: 2024-12-15

Figure 10: Funding & Capital Opportunities listing grants, subsidies and micro-loans.

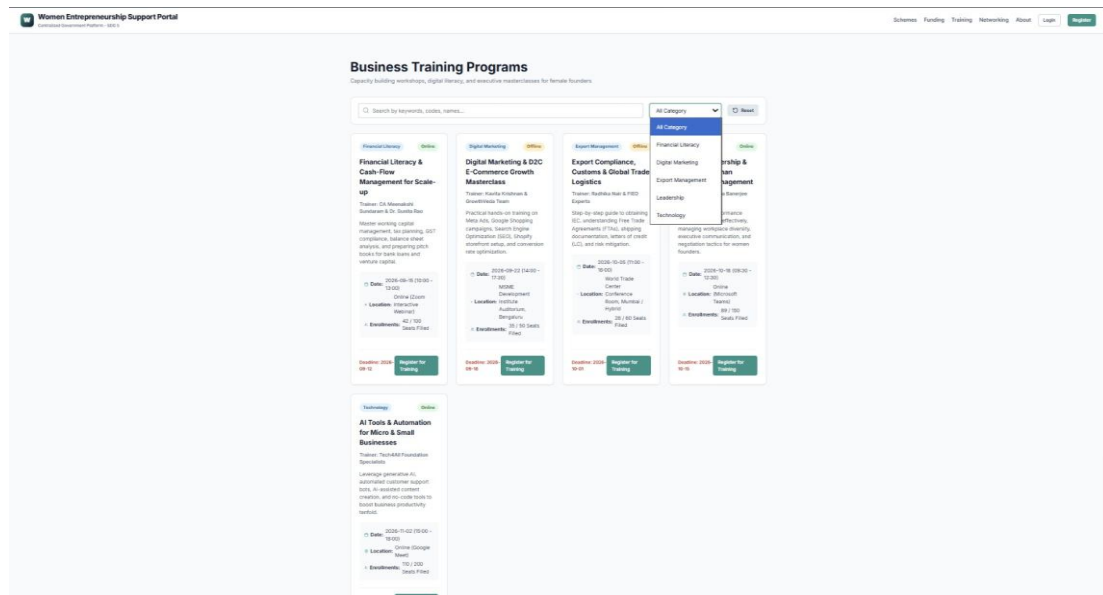


Figure 11: Business Training Programs, filterable by category.

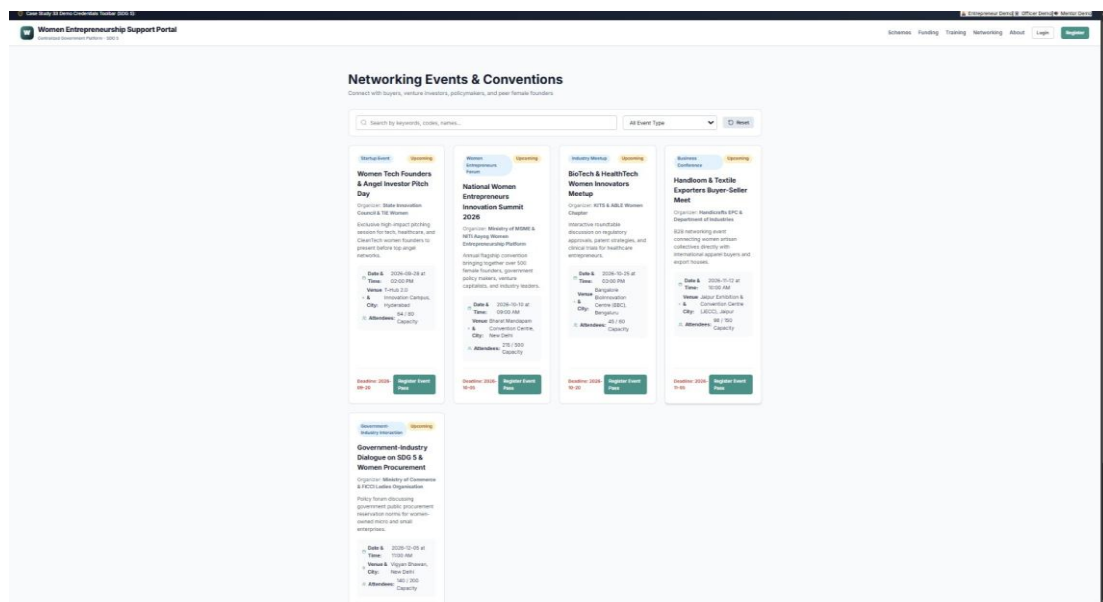


Figure 12: Networking Events & Conventions.

10. Challenges Encountered

- **Role-Based Access Design:** separating Entrepreneur, Mentor and Officer views into distinct protected route trees while sharing common components (Navbar, Sidebar, Modal) required careful structuring of the client/src/pages directory and the AuthContext/ProtectedRoute logic.
- **Sector-Based Recommendation Logic:** matching recommended schemes, funding and mentors to an entrepreneur's declared business sector required designing consistent sector taxonomies across the Schemes, Funding and Mentor Directory modules.
- **Git Repository Setup:** verifying that .gitignore correctly excluded node_modules and other build artefacts before the initial commit required an explicit git check-ignore check.
- **Docker Environment Availability:** as with prior coursework, running Docker Desktop locally has constraints on the development machine, so the Docker Compose design for this project has been prepared and reviewed but not yet executed end-to-end.

- Testing and Integration: coordinating testing across registration, business verification, applications, training, events and mentorship requires a structured test plan, and formal execution of these test cases remains pending.

11. Future Work and Improvements

- Automated eligibility matching between an entrepreneur's business profile and scheme/funding criteria.
- Email or SMS notifications for application status changes and upcoming training/event deadlines.
- Document upload and verification workflow for Udyam registration and supporting KYC documents.
- Analytics and reporting dashboard for officers covering scheme uptake, funding disbursed and training participation by sector and region.
- A companion mobile application for entrepreneurs in areas with limited desktop access.
- Automated testing (unit and integration) to improve reliability and reduce regression risk.
- A CI/CD pipeline to automate build, test and deployment once containerised deployment is verified.
- Expansion of the mentorship module with scheduling and video-call integration for mentoring sessions.

12. Conclusion

The Women Entrepreneurship Support Portal demonstrates how a structured, government-backed web platform can replace the current fragmented landscape of scheme, funding, training and mentorship information with a single, centralised alternative. By combining a React (Vite) frontend, a Node.js/Express backend and a MySQL database within a layered, role-based architecture, and by applying sound software engineering practice through Git version control and a containerised Docker deployment design, the project addresses the core inefficiency identified in the problem statement: fragmentation of support information across multiple websites. In doing so, the portal supports the aims of SDG 5 by helping to make government support for women entrepreneurs more discoverable, traceable and accessible. While branching, merge history and containerised deployment are still to be demonstrated end-to-end, and formal test execution remains pending, the initial repository, application structure and Docker design establish a solid foundation for the enhancements outlined above.

13. References

- React Documentation. Available at: <https://react.dev/>
- Vite Documentation. Available at: <https://vite.dev/>
- Node.js Documentation. Available at: <https://nodejs.org/en/docs>
- Express Documentation. Available at: <https://expressjs.com/>
- MySQL Documentation. Available at: <https://dev.mysql.com/doc/>
- Docker Documentation. Available at: <https://docs.docker.com/>
- Docker Compose Documentation. Available at: <https://docs.docker.com/compose/>
- Git Documentation. Available at: <https://git-scm.com/doc>
- United Nations. Sustainable Development Goal 5: Gender Equality. Available at: <https://sdgs.un.org/goals/goal5>

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Analysis & Project Design(20%)	Clearly explains the problem statement, objectives, requirements, expected outcomes, and well-organized project structure with proper justification.	Good analysis and project structure with minor omissions.	Basic analysis and repository structure with limited justification.	Incomplete or unclear analysis and project organization.
Git Repository & Branching Strategy(20%)	Repository initialized correctly with appropriate branching strategy, clear workflow, and proper repository organization.	Repository and branching strategy are mostly correct.	Basic Git setup with limited branching implementation.	Incorrect or incomplete Git repository and branching strategy.
Version Control Workflow & Commit History(10%)	Demonstrates meaningful commits, branching, merging, synchronization, conflict resolution, and professional commit messages with clear history.	Most Git operations demonstrated correctly with good commit history.	Limited Git workflow and commit practices.	Poor workflow and inadequate commit history.
Docker Implementation(20%)	Well-designed Docker environment, Dockerfile, and Docker Compose configuration with clear explanation of services, networking, volumes, and dependencies.	Functional Docker implementation with minor issues.	Basic Docker implementation with limited explanation.	Incomplete or incorrect Docker implementation.
Deployment, Testing & Results(15%)	Application successfully deployed and tested with appropriate screenshots and evidence of successful execution.	Deployment completed with minor issues.	Partial deployment and testing demonstrated.	Deployment unsuccessful or insufficient evidence.
Documentation, Challenges & Future Enhancements(15%)	Professional report (10–15 pages) with diagrams, screenshots, references, discussion of benefits, challenges, and future improvements.	Good documentation with minor omissions.	Basic documentation with limited discussion.	Poor documentation and insufficient analysis.

Criteria	Mark
System Requirement Analysis (30%)	/5
Selection of Software Architecture (25%)	/5
Architecture Diagram and Component Design (20%)	/5
Component Interaction and Quality Attribute Analysis (15%)	/5
Architectural Justification and Technical Presentation (10%)	/5
TOTAL	/25